

Soft Spaces in Low-Qubit Hilbert Spaces – Phase 3.1

Retrospective Whitepaper:
Basis-Invariant Projector Reformulation, REAL–NULL Spectral Structure,
Independent Replication, and Dimensional Confirmation

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Abstract

Phase 3.1 was initiated after the Phase-2 audit exposed a decisive representation problem: a hotspot statistic based on an arbitrarily selected two-eigenvector slice inside a larger exactly degenerate eigenspace could change under a legal unitary rotation while the Hamiltonian and its spectrum remained unchanged. Phase 3.1 therefore reformulated the Soft Spaces programme around complete spectral projectors and explicitly separated representation invariance, Hamiltonian degeneracy, local prominence, spectral $\pm E$ organization, REAL–NULL decomposition, independent replication, and cross-dimensional replication.

The central operator is the regularized second-order projector response

$$K_P^{(\eta)}(E) = PVQ \frac{E - QHQ}{(E - QHQ)^2 + \eta^2} QVP, \quad Q = \mathbb{I} - P,$$

with a normalized Frobenius response defined entirely on complete exact eigenspaces. This construction passed direct basis-rotation invariance tests to numerical precision. A subsequent Hamiltonian audit showed that the original sparse five-term random-Pauli model was massively degenerate; an eleven-term working model reduced this degeneracy sufficiently to expose non-trivial complete-eigenspace structure without returning to arbitrary eigenvector-pair selection.

The principal empirical finding of Phase 3.1 is not an isolated hotspot coordinate. It is a strong $E \leftrightarrow -E$ organization shared by REAL and spectrum-matched Haar NULL models, together with a smaller positive REAL-specific excess. The excess was first detected across the original 8Q development seeds, then independently replicated on a frozen new 8Q seed block under five predeclared acceptance gates, and finally confirmed at both 7Q and 9Q under the same gate structure. The dimensional results suggest that REAL remains close to perfect $\pm E$ organization while NULL increasingly approaches the same organization as Hilbert-space dimension grows, causing the contrast $\Delta r = r_{\text{REAL}} - r_{\text{NULL}}$ to decrease from 7Q to 9Q.

This whitepaper records Phase 3.1 as a completed methodological and empirical milestone. It distinguishes what was falsified, what was repaired, what was replicated, what remains only suggestive, and which questions are deliberately transferred to Phase 3.2. It is not intended to replace the technical Phase-3 article; rather, it provides a compact research map and provenance record for the completed Phase-3.1 programme.

Status: completed retrospective milestone document. Phase 3.1 is treated as closed except for reproducibility maintenance and editorial corrections.

Repository convention: the Phase-3.1 repository series is named `v31_0–v31_8`. Historical internal computational labels and frozen RNG namespaces such as `v30.2` are retained where they record the original execution provenance.

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1 Purpose of this whitepaper

A conventional article emphasizes the final argument. A research whitepaper serves a different purpose: it preserves the decision path. Phase 3.1 is particularly suited to such documentation because the most important progress came from successive falsification steps rather than from a single monotonic search for stronger effects.

The programme can be summarized as the following transition:

basis-dependent pair hotspot $\longrightarrow$ complete projector response $\longrightarrow$ shared spectral symmetry $\longrightarrow$ small replica

The objective of this document is therefore fourfold:

1. record the methodological failure that motivated Phase 3.1;
2. state the repaired basis-invariant mathematical object;
3. summarize the evidence sequence from v31_0 through v31_8;
4. define the boundary between established Phase-3.1 results and Phase-3.2 hypotheses.

2 The Phase-2 failure that Phase 3.1 had to solve

Suppose an eigenvalue E has multiplicity $d > 1$ with projector

$$P_E = \sum_{a=1}^d |E, a\rangle \langle E, a|.$$

Any unitary transformation $U \in U(d)$ generates an equally valid orthonormal basis of the same physical eigenspace. A statistic attached to a selected subset of basis vectors can therefore be physically meaningful only if it is invariant under such rotations, or if an independent physical principle selects a privileged basis.

The Phase-2 pair-based construction did not satisfy this requirement in the final degeneracy audit. In schematic form, if

$$(|E, 1\rangle, |E, 2\rangle) \mapsto (|E, 1'\rangle, |E, 2'\rangle)$$

under a legal unitary rotation within the same degenerate eigenspace and the hotspot sign changes, then the pair score is not a function of H and V alone.

This was not a small numerical nuisance. It changed the ontological status of the candidate object. Phase 3.1 therefore adopted the rule:

The primary object must be a complete degeneracy-closed subspace, represented by its projector, unless a separate physical basis-selection mechanism is demonstrated.

3 Projector reformulation

Let the Hermitian Hamiltonian have spectral resolution

$$H = \sum_g E_g P_g, \quad P_g P_h = \delta_{gh} P_g, \quad \sum_g P_g = \mathbb{I}. \quad (1)$$

For a target exact eigenspace $P = P_j$, define

$$Q = \mathbb{I} - P.$$

The decomposition

$$\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H}$$

is purely a Hilbert-space partition at this stage; Q is not yet interpreted as an external environment. For perturbation V , reference energy E , and regularization $\eta > 0$, Phase 3.1 uses

$$K_P^{(\eta)}(E) = PVQ \frac{E - QHQ}{(E - QHQ)^2 + \eta^2} QVP. \quad (2)$$

This is a regularized second-order effective operator of Feshbach/resolvent type. Using the exact eigenspace decomposition outside P ,

$$K_P^{(\eta)}(E_j) = \sum_{g \neq j} r_{jg} PVP_gVP, \quad (3)$$

where

$$r_{jg} = \frac{E_j - E_g}{(E_j - E_g)^2 + \eta^2}. \quad (4)$$

Define

$$A_g = PVP_gVP.$$

The normalized projector response is

$$C(P, V) = \frac{\left\| \sum_{g \neq j} r_{jg} A_g \right\|_F}{\sum_{g \neq j} |r_{jg}| \|A_g\|_F}. \quad (5)$$

The numerator measures the norm of the coherent second-order sum; the denominator normalizes by the summed magnitudes of the individual virtual channels. Low values therefore correspond to stronger cancellation of channel contributions, although a complete physical interpretation of that cancellation is deliberately deferred.

4 Why the new score is basis invariant

Let U_P rotate the basis inside the target eigenspace and let U_g rotate inside a complementary exact eigenspace. The projectors themselves remain unchanged. In a target-space matrix representation,

$$A_g \mapsto U_P^\dagger A_g U_P.$$

Because the Frobenius norm is unitarily invariant,

$$\|U_P^\dagger A_g U_P\|_F = \|A_g\|_F.$$

The same transformation applies to the coherent sum in the numerator of Eq. (5). Hence

$$C(P, V) = C(U_P P U_P^\dagger, V) \quad (6)$$

for legal basis changes internal to the exact eigenspaces.

The v31_0 invariance audit tested precisely this requirement by Haar rotating each exactly degenerate basis while leaving H and the spectrum unchanged. Variations were at numerical round-off scale, approximately 10^{-16} in the smoke test, with Hamiltonian reconstruction errors at approximately 10^{-15} – 10^{-14} . The invariance gate passed.

This is the first hard boundary of Phase 3.1: the previous pair-based representation dependence was removed.

5 Hamiltonian degeneracy audit

The basis repair immediately exposed a second issue. In the original 8Q five-term random-Pauli ensemble, the spectrum contained only four exact energy groups of size 64. Once complete eigenspaces were treated correctly, the earlier local pair-prominence structure collapsed toward zero.

The `v31_1` scan therefore varied the number of random Pauli terms while holding the conceptual scoring framework fixed. A representative 8Q seed showed the progression

$$5 \text{ terms} : 4 \times 64, \quad 7 \text{ terms} : 16 \times 16, \quad 9 \text{ terms} : 16 \times 16, \quad 11 \text{ terms} : 64 \times 4.$$

The purpose was not to tune toward a desired hotspot. It was to reduce massive spectator degeneracy while preserving a sparse, interpretable Pauli Hamiltonian.

A crucial decision followed: the old near-neighbour pair-eligibility condition was not relaxed post hoc merely to restore candidate pairs. Instead, `v31_2` removed the pair gate entirely and scanned complete exact eigenspaces.

6 Complete-eigenspace prominence and the emergence of $\pm E$ structure

For each target eigenspace and perturbation family f , define a REAL–NULL contrast

$$\Delta C_f(j) = C_{\text{REAL}}(P_j, V_f) - C_{\text{NULL}}(P_j, V_f).$$

A conservative family-combined response is

$$R_j = \min_f \Delta C_f(j), \tag{7}$$

where the frozen families are dephasing-like Z/ZZ and transverse X/XX perturbations.

Local prominence is defined against neighbouring eigenspaces in energy order:

$$\Pi_j = R_j - \text{median}\{R_k : k \in \mathcal{N}_j\}, \tag{8}$$

with a frozen local radius of five eigenspaces in the primary scan.

At 8Q with eleven Pauli terms, `v31_2` produced non-trivial local prominences across all 12 development seeds. For seed 25042000 the top prominence was

$$\Pi_{\max} = +0.632689923.$$

More important than the maximum itself was a conspicuous pattern: high-prominence structures repeatedly occurred in energy-reflected pairs E and $-E$.

This observation changed the target of inference. A scan necessarily has a maximum; therefore “a large maximum exists” is not by itself a significance claim. Phase 3.1 shifted attention from isolated maxima toward organization under the transformation

$$E \mapsto -E.$$

7 Frozen significance tests and the `v31_3/v31_4` correction

The first significance script introduced shuffle tests for top prominence and $\pm E$ pairing. During validation, however, a freeze defect was discovered: the implementation used the current version

string inside deterministic hash namespaces for NULL and perturbation generation. Changing the version label therefore changed the underlying realization.

This was a methodological failure of provenance, not a failure of the physical hypothesis. It was repaired immediately. The corrected `v31_4` implementation froze the physical RNG namespace exactly to the earlier discovery model, historically labelled `v30.2`.

The corrected 12-seed significance analysis showed:

top-prominence significance:	0/12 seeds at $p < 0.05$,
$\text{median}(p_{\text{top}})$	≈ 0.637 ,
$\text{median}(r_{\pm E})$	≈ 0.963 ,
p_{corr}	≈ 0.000500 for all 12 seeds,
$p_{\text{pair}} < 0.05$	for all 12 seeds.

Thus the isolated hotspot height was not statistically exceptional under the frozen shuffle baseline, whereas the $\pm E$ organization was extremely strong. This distinction is central to the mature Phase-3.1 interpretation.

8 REAL–NULL symmetry decomposition

A strong $\pm E$ pattern can arise from the algebraic structure of the Hamiltonian ensemble itself. Phase 3.1 therefore asked whether the pairing belonged specifically to REAL or was largely inherited by the spectrum-matched NULL.

For each seed define

$$r_{\text{REAL}} = \text{Corr}[\Pi_{\text{REAL}}(E), \Pi_{\text{REAL}}(-E)], \quad (9)$$

$$r_{\text{NULL}} = \text{Corr}[\Pi_{\text{NULL}}(E), \Pi_{\text{NULL}}(-E)], \quad (10)$$

$$\Delta r = r_{\text{REAL}} - r_{\text{NULL}}. \quad (11)$$

The `v31_5` decomposition showed that both components are highly symmetric. Across the original 12 8Q seeds,

$$\text{median}(r_{\text{REAL}}) = 1.000000, \quad \text{median}(r_{\text{NULL}}) \approx 0.967896,$$

with

$$\text{median}(\Delta r) \approx +0.032104.$$

Eleven of twelve seeds had positive Δr , although only one seed individually passed a conventional seedwise permutation threshold. This is exactly the situation for which across-seed inference is more appropriate than isolated seedwise significance.

The correct qualitative statement became:

Most of the observed $\pm E$ organization is structural and shared with NULL, but REAL carries a smaller, unusually consistent positive excess.

9 Across-seed inference at 8Q

The v31_6 analysis introduced no new physical simulation. It treated the frozen 12 values of Δr as the statistical endpoint. The results were

$$\begin{aligned}
 \overline{\Delta r} &= +0.050466, \\
 \text{median}(\Delta r) &= +0.032104, \\
 11/12 \text{ seeds positive}, \\
 p_{\text{sign},1s} &= 0.00317383, \\
 p_{\text{Wilcoxon},1s} &= 0.00146449, \\
 95\% \text{ CI}_{\text{boot}}(\overline{\Delta r}) &= [0.022297, 0.084095], \\
 95\% \text{ CI}_{\text{boot}}(\text{median } \Delta r) &= [0.012663, 0.063867], \\
 d_z &= +0.878293.
 \end{aligned}$$

Because r_{REAL} often saturated numerically near one, Fisher- z transformations were treated only as secondary diagnostics. The primary inference rested on sign consistency, Wilcoxon signed ranks, bootstrap intervals, and the directly interpretable difference Δr .

10 Independent frozen replication: v31_7

A development-set effect should not be promoted to a replicated result without a new seed block and frozen acceptance criteria. v31_7 therefore used independent Hamiltonian seeds 25043000–25043011 and predeclared five simultaneous gates:

- C1. median $\Delta r > 0$;
- C2. at least 9 of 12 seeds have $\Delta r > 0$;
- C3. one-sided exact sign-test $p < 0.05$;
- C4. one-sided Wilcoxon signed-rank $p < 0.05$;
- C5. the 95% bootstrap confidence interval for median Δr lies entirely above zero.

All five gates passed. The independent 8Q replication produced

$$\begin{aligned}
 11/12 \text{ positive seeds}, \\
 \overline{\Delta r} &= +0.048107, \\
 \text{median}(\Delta r) &= +0.040406, \\
 p_{\text{sign},1s} &= 0.00317383, \\
 p_{\text{Wilcoxon},1s} &= 0.00097632, \\
 95\% \text{ CI}_{\text{boot}}(\text{median } \Delta r) &= [0.024192, 0.056815], \\
 d_z &= +1.056863.
 \end{aligned}$$

This is the first point at which the REAL-specific excess deserves the word *replicated* within the Phase-3 programme.

11 Cross-dimensional confirmation: v31_8

The next question was whether the replicated 8Q endpoint generalized beyond the dimension in which it had been developed. v31_8 therefore applied the same conceptual kernel and the same five gates separately to 7Q and 9Q using a new seed block 25044000–25044011.

Table 1: Primary dimensional replication results.

Dimension	Mean Δr	Median Δr	Positive seeds
7Q	0.191973	0.183939	12/12
8Q independent replication	0.048107	0.040406	11/12
9Q	0.015577	0.010472	12/12

At 7Q,

$$\begin{aligned}
 p_{\text{sign},1s} &= 0.00024414, \\
 p_{\text{Wilcoxon},1s} &= 0.00048816, \\
 95\% \text{ CI}_{\text{boot}}(\text{median } \Delta r) &= [0.080120, 0.244121], \\
 d_z &= +1.342567.
 \end{aligned}$$

At 9Q,

$$\begin{aligned}
 p_{\text{sign},1s} &= 0.00024414, \\
 p_{\text{Wilcoxon},1s} &= 0.00048816, \\
 95\% \text{ CI}_{\text{boot}}(\text{median } \Delta r) &= [0.003263, 0.019466], \\
 d_z &= +0.796624.
 \end{aligned}$$

Both dimensions passed all five predeclared gates. The formal conclusion was therefore *confirmed across dimensions* for the tested 7Q–9Q range.

12 The dimensional clue

The dimensional result contains more information than a simple yes/no replication. The mean correlations were approximately

Table 2: Mean $\pm E$ correlation by dimension.

Dimension	r_{REAL}	r_{NULL}	Δr
7Q	0.984588	0.792616	0.191973
8Q	0.998061	0.949954	0.048107
9Q	0.999795	0.984218	0.015577

The most economical empirical description is therefore not that REAL loses its organization with increasing dimension. Rather,

$$r_{\text{REAL}}(Q) \approx 1, \quad r_{\text{NULL}}(Q) \uparrow 1, \quad \Delta r(Q) \downarrow \quad (12)$$

for the observed 7Q–9Q range.

Equation (12) is a hypothesis-generating pattern, not a scaling law. Three dimensions are insufficient to establish functional dependence. Nevertheless it provides a concrete mechanism question for Phase 3.2: why does the NULL construction “catch up” to REAL so rapidly as dimension grows?

A particularly important observation is the 7Q seed 25044004. That case had 128 exact groups of size one – a fully nondegenerate spectrum – yet produced approximately

$$r_{\text{REAL}} = 0.985464, \quad r_{\text{NULL}} = 0.473394, \quad \Delta r = +0.512070.$$

Thus the surviving REAL-specific effect does not require exact degeneracy. This sharply separates the new Phase-3.1 endpoint from the old basis-degeneracy artifact.

13 The Phase-3.1 working equation

The complete logic of the phase can be written as an operator-to-statistic chain:

$$H, V \longrightarrow P \longrightarrow K_P^{(\eta)}(E) \longrightarrow C(P, V) \longrightarrow \Delta C(P) \longrightarrow \Delta r.$$

A compact phenomenological endpoint is

$$\boxed{\mathcal{S}_{\text{SS}} = \text{Corr}_{E,-E}[\Pi_{\text{REAL}}(E), \Pi_{\text{REAL}}(-E)] - \text{Corr}_{E,-E}[\Pi_{\text{NULL}}(E), \Pi_{\text{NULL}}(-E)]}. \quad (13)$$

Within the Phase-3.1 implementation this is essentially the replicated quantity Δr .

Equation (13) should not be presented as a fundamental physical law. It is a working observable that summarizes the experimentally useful residual after a dominant shared structural symmetry has been subtracted.

14 What Phase 3.1 does and does not establish

14.1 Established within the computational model

The evidence supports the following statements:

1. the original arbitrary pair score fails a basis-invariance requirement inside larger exact degeneracies;
2. a complete-projector formulation removes that defect to numerical precision;
3. the five-term Hamiltonian is excessively degenerate for the intended subspace analysis;
4. an eleven-term working model yields non-trivial complete-eigenspace prominence structure;
5. the dominant $\pm E$ organization is largely shared by REAL and NULL;
6. a smaller REAL-specific excess remains after this confound is separated;
7. that excess is reproducible across seeds, independently replicated at 8Q, and confirmed at 7Q and 9Q under predeclared gates;
8. the residual effect can exist without exact degeneracy.

14.2 Not established

Phase 3.1 does not establish:

1. a universal physical law called “Soft Spaces”;
2. a unique functional scaling law for $\Delta r(Q)$;
3. an environmental or dissipative interpretation of Q ;
4. that the $\pm E$ symmetry is specifically caused by open-system physics;
5. that the random-Pauli ensemble is representative of arbitrary physical Hamiltonians;
6. that REAL-specific spectral organization automatically implies protection of a quantum state or logical qubit.

This separation is intentional. The strongest outcome of Phase 3.1 is a narrower and more defensible object than the initial hotspot hypothesis.

15 Methodological lessons

Phase 3.1 produced several process lessons that should remain binding in later phases.

Representation invariance before interpretation. A visually stable or statistically significant feature is not sufficient if it depends on an arbitrary eigenbasis convention.

Freeze semantics, not only seeds. Deterministic random generation can change when version labels enter hash namespaces. Provenance therefore includes the full seed namespace, not merely the integer Hamiltonian seed.

Separate discovery from confirmation. The same seed block should not be used both to notice a pattern and to claim independent replication.

Distinguish structural background from REAL-specific residual. A strong pattern present in both REAL and NULL is not evidence for the distinguishing hypothesis by itself.

Do not rescue a failed object by post-hoc threshold changes. When pair eligibility ceased to be meaningful after the degeneracy audit, the correct response was to redefine the physical object, not loosen epsilon.

16 Version map and provenance

Repository version	Scientific role
v31_0	Projector-score basis-invariance audit.
v31_1	Hamiltonian term-count and degeneracy diagnostic.
v31_2	Complete exact-eigenspace hotspot/prominence scan.
v31_3	Initial significance/pairing test; later identified as not fully frozen because version text entered RNG namespaces.
v31_4	Corrected frozen significance and $\pm E$ pairing test, preserving the historical v30.2 physical namespace.
v31_5	REAL/NULL $\pm E$ symmetry decomposition and seedwise Δr .
v31_6	Across-seed statistical inference on frozen 8Q development values.

Repository version	Scientific role
v31_7	Independent predeclared 8Q replication.
v31_8	Predeclared 7Q/9Q dimensional replication.

The repository rename from historical `v30_*` filenames to `v31_*` is organizational: it aligns file naming with Phase 3.1. Raw output text and computational namespaces may still contain historical `v30.x` labels where they document the exact code state under which results were generated. Those strings should not be globally rewritten when they participate in deterministic RNG or provenance.

17 Boundary to Phase 3.2

Phase 3.1 ends with an empirical and methodological fact pattern. Phase 3.2 begins with a mechanism question.

The primary next hypothesis is that the near-perfect $\pm E$ structure may arise from an exact or approximate anticommuting symmetry operator S satisfying

$$SHS^{-1} \approx -H, \quad \text{or} \quad \{S, H\} \approx 0. \quad (14)$$

If such a symmetry exists, it can explain the spectral pairing algebraically before any open-system interpretation is invoked.

Only after the symmetry mechanism is understood should Eq. (2) be re-examined as a possible self-energy or effective-coupling object. The projection structure resembles Feshbach and projection-operator methods, but in Phase 3.1 Q is simply the complementary subspace. A genuine open-quantum-system statement would require additional ingredients such as a physically distinguished environment, reduced dynamics, decoherence, dissipation, or an equivalent operational interpretation.

Thus the formal handoff is

Phase 3.1: establish invariant replicated structure $\implies$ Phase 3.2: identify its algebraic mechanism.

18 Final Phase-3.1 conclusion

Phase 3.1 did not validate the original conception of a Soft Space as an isolated, ordinal pair of eigenvectors with an exceptional local score. It did something more useful: it removed a representation artifact, exposed excessive Hamiltonian degeneracy, rebuilt the score on complete spectral projectors, separated a dominant structural $\pm E$ symmetry from the REAL-specific component, and then subjected that residual to development-set inference, independent replication, and dimensional confirmation.

The strongest defensible endpoint is therefore:

Within the frozen low-qubit random-Pauli model, complete spectral subspaces exhibit a strong $E \leftrightarrow -E$ organization that is largely structural and shared by REAL and spectrum-matched NULL constructions. On top of this background, a smaller positive REAL-specific correlation excess remains. This excess is basis-invariant by construction, independently replicated at 8Q, confirmed under the same predeclared gate structure at 7Q and 9Q, and demonstrably not dependent on exact degeneracy.

The scientific value of the result lies as much in its narrowing as in its confirmation. Phase 3.1 converts a broad hotspot hypothesis into a smaller, reproducible operator-level phenomenon. That is an appropriate point at which to close the phase and begin the algebraic mechanism programme of Phase 3.2.

Document note

This whitepaper is retrospective. Numerical values are recorded from the completed Phase-3.1 computational sequence. It is intended as a project roadmap, provenance record, and conceptual bridge to the Phase-3.2 whitepaper and does not supersede raw output files or the technical Phase-3 article.